

Whisking Towards a Greener Future: Revamping Electric Whisk for Eco-Friendly Innovation

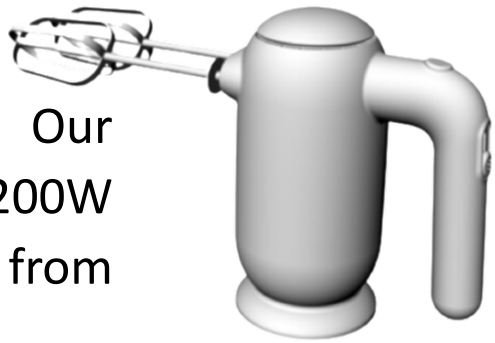


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Introduction

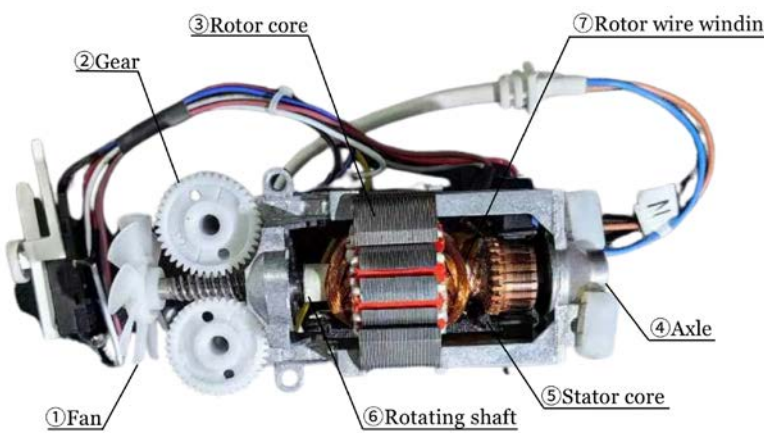
An electric whisk, a kind of small handheld device used to whip ingredients are widely used in home kitchens and small workshops. Our competitor product the SUPOR electric whisk, due to its lightweight, 200W enough power and stir bar push-out button design, has positive feedback from the customers. However, shortages still exist in SUPOR electric whisk[1].



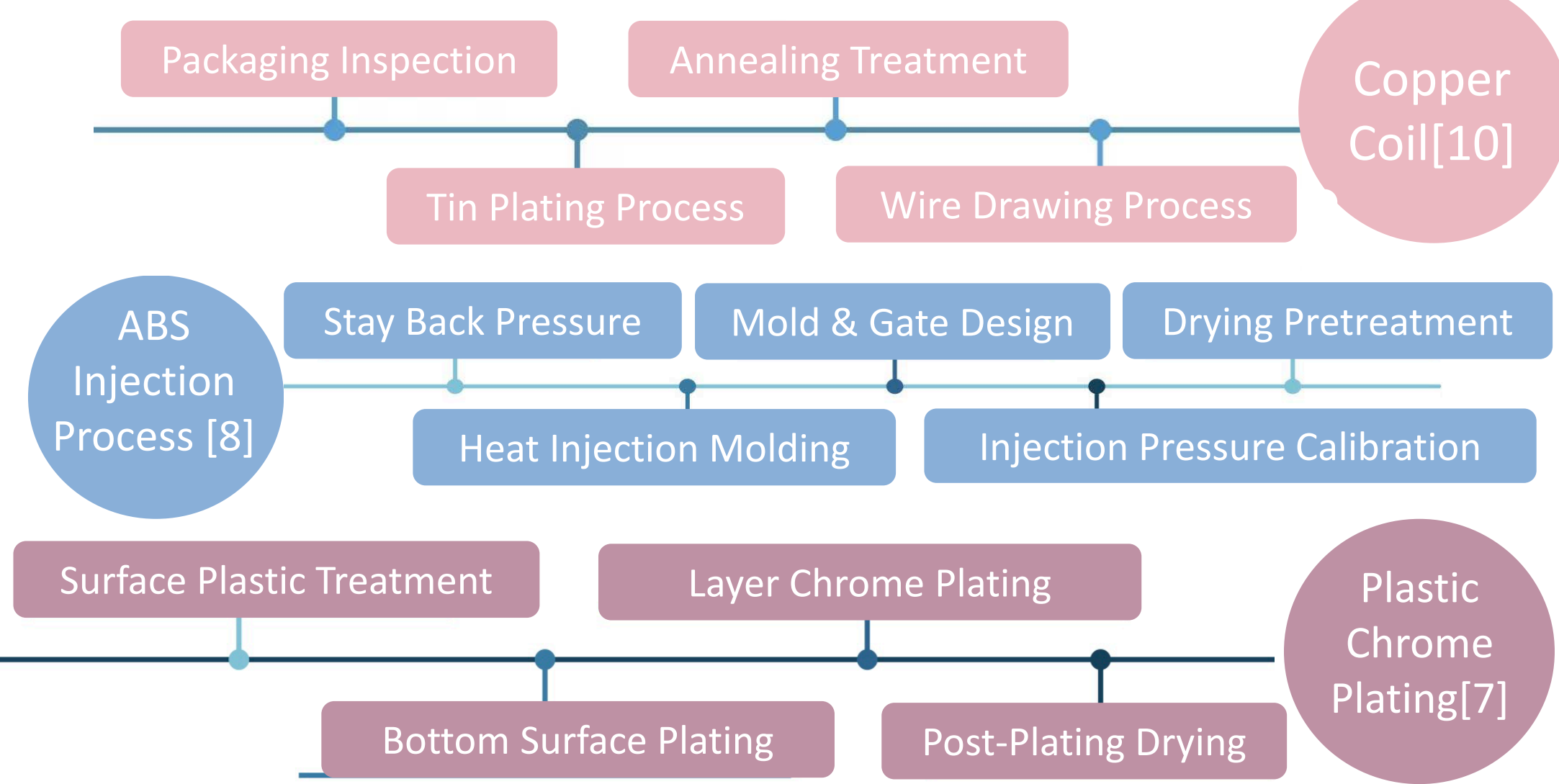
User requirement

User Requirement	Design Input	Out Put	Project Goal
Long Service Life	Heat resistance	ABS plastic shell	For the shell, we are about to enhance heat dissipation capability.
	Improved shell shape	Longer sustaining time	
	Copper wire	Good conductivity	
Environmentally Friendly	Energy saving	Soft magnetism	For the engine, a new material will be applied to save energy.
	Process	Larger magnetic flux	
	Environmental Protection	Smaller electrical losses	
Environmentally Friendly	Traditional chrome plating	Durable, safe, and resistant to rusting	For the rods, the coating process can be optimized for eco-friendliness.
	Plastic chrome plating	More affordable	
		environmentally friendly	

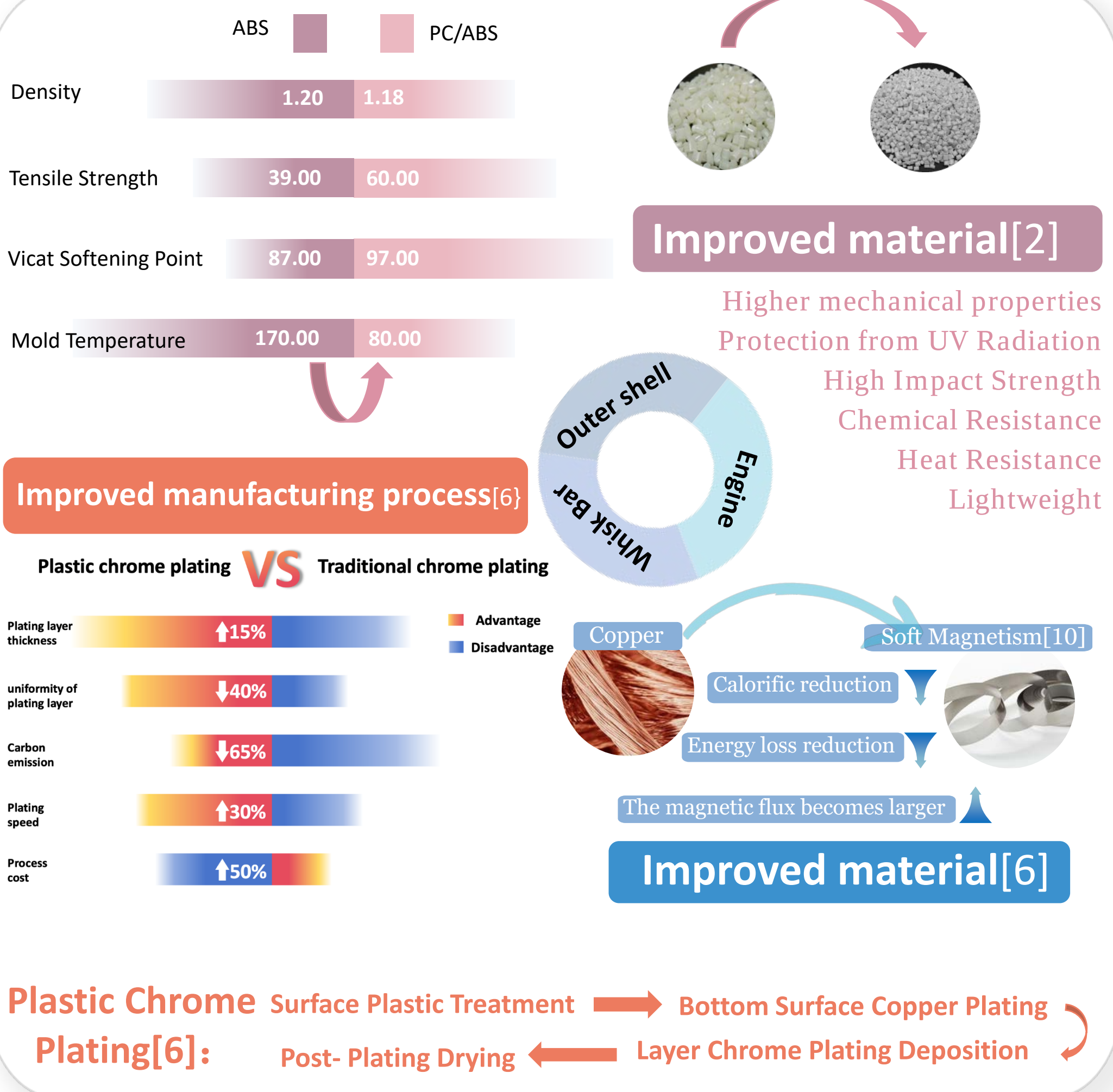
Material & Property & Failure Mode

	Figure	Material	Property	Failure mode
Outer shell		ABS[2][3]	Excellent mechanical properties: 1. High impact strength 2. Dimensional stability 3. Electrical insulation 4. Processability	High temperatures cause it to deform or lose its mechanical properties
Engine		① Fan	④ Axle	
		② Gear	⑤ Stator core	
		③ Rotor core	⑥ Rotating shaft	
Whisk bar		③ Stainless Steel[9]	Excellent mechanical properties: 1. High strength 2. Corrosion resistance	1. Tensile stress : Crack propagation welding failure 2. Temperature and humidity: Intergranular / crevice corrosion
		④ Copper[6]	1. High electrical conductivity 2. Malleability 3. Corrosion resistance	1. Tensile strength: than 275 MPa 2. Elongation 610 and 65 are 5% and 10%
		Chrome plating[7]	1. Good corrosion resistance 2. Excellent wear resistance 3. Strong heat resistance	High-temperature steam environment, humid environment, pH change, chemical corrosion
Whisk bar		Q195 food grade carbon steel	Malleability, weldability, rotting resistance, and plasticity	Tensile strength: 315-430 MPa, Elongation: ≥ 33%, Yield strength: 195 Mpa.

Manufacture



Improvement



Sustainable

For the shell, the service life of the PC/ABS shell is generally 15 years[3], which is greatly extended[4]. By using the LCA method, the evaluation results show that the predicted carbon footprint is **reduced by 8% compared to ABS**[4].

For the material of the wire, soft magnetic materials can increase magnetic flux, convert more electrical energy into magnetic energy, and then convert it into mechanical energy, thereby reducing the generation and **loss of thermal energy** and achieving the goal of energy conservation and environmental protection[6][10].

For the rods, the metal surface is covered with a thin layer of special plastic followed by chrome plating, bringing significant data improvements. The coating thickness is increased by 15%, the surface uniformity variance is reduced by 40%, and the product quality is greatly improved. At the same time, **the carbon footprint has been reduced by 65%**, showing a high degree of environmental protection [7].

Reference

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